FDA RoLoRA Dual Track

Notation, Symbols, and Hyperparameters

- r: Current global communication round.
- $\mathcal{M}^{(r)}$: The active matrix for round r (either A or B).
- e: Current local epoch iteration counter.
- e^*: Final realized local epoch count.
- $\Delta_i^{(e)}$: Raw subspace parametric drift for client i.
- $Var_{subspace}(\Delta^{(e)})$: The variance calculated directly and exclusively on the low-rank subspace coordinates (either matrix A or B) without mapping to the expanded space.
- Unique Hyperparameters:
  - $e_{min}, e_{max} \in \mathbb{N}$: Minimum and maximum local epoch bounds.
  - $\theta_A, \theta_B \in \mathbb{R}^+$: Independent matrix specific variance thresholds. θ_A only updates during rounds where $\mathcal{M}^{(r)} = A$, preventing geometric contamination.
  - $\theta_{current}^{(r)}$: The loaded active threshold mapping to the current active matrix.
  - $\rho \in (0, 1)$: Smoothing factor for Exponential Moving Average (EMA) threshold update.
  - $\alpha_{ema} \in \mathbb{R}^+$: Constant expansion coefficient utilized exclusively for the Pure EMA variant.
  - $K \in \mathbb{N}$: Window size for Simple Moving Average (SMA) threshold update.

1) Dual Track Initialization and Subspace Toggling At round r, $\mathcal{M}^{(r)} \in \{A, B\}$ is determined. The system fetches the specific persistent threshold corresponding exclusively to the active matrix to govern the synchronization dynamics of the current round.

$$\theta_{current}^{(r)} = \begin{cases} \theta_A & \text{if } \mathcal{M}^{(r)} = A \\ \theta_B & \text{if } \mathcal{M}^{(r)} = B \end{cases}$$

$$W_{i,0}^{(r)} = W_{act}^{(r)} \quad \forall i \in S_r$$

2) Local Training and Direct Subspace Variance Clients advance using independent SGD steps:
$W_{i,t}^{(r)} = W_{i,t-1}^{(r)} - \eta_c \nabla \mathcal{L}_i(W_{i,t-1}^{(r)})$. During evaluation epochs $e \geq e_{min}$, raw subspace drifts are measured. Instead of projection or linear approximation, exact variance is computed directly on the isolated low-rank subspace.

$$\Delta_i^{(e)} = W_{i,e}^{(r)} - W_{act}^{(r)}$$

$$Var_{subspace}(\Delta^{(e)}) = \frac{1}{|S_r|} \sum_{i \in S_r} \left\| \Delta_i^{(e)} \right\|_2^2 - \left\| \frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e)} \right\|_2^2$$

3) Independent Synchronization Halt The halting mechanism restricts computation based on the isolated $\theta_{current}^{(r)}$, ensuring asymmetrical convergence geometries between projection (A) and expansion (B) matrices do not prematurely cross contaminate the stopping criteria.

$$\text{Break Local Training if } Var_{subspace}(\Delta^{(e)}) > \theta_{current}^{(r)} \quad \vee \quad e \geq e_{max}$$

4) Global Aggregation and Independent Threshold Evolution Following aggregation of the subspace drifts ($\nabla_{pseudo}^{(r)} = -\frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e^*)}$), the server optimization step updates the active parameters. Crucially, the threshold for $\mathcal{M}^{(r)}$ mutates independently, preserving state for its next cycle. The threshold for the inactive matrix remains frozen.
Let the epoch scaled proposed target threshold be defined as:

$$\theta_{target}^{(r)} = \left(\frac{e_{max}/2}{e^*} \right) Var_{subspace}(\Delta^{(e^*)})$$

The threshold for the subsequent active cycle, $\theta_{\mathcal{M}^{(r)}}^{(next)}$, is updated using one of the following variants:
Standard Variant:

$$\theta_{\mathcal{M}^{(r)}}^{(next)} = \theta_{target}^{(r)}$$

Simple Moving Average (SMA) Variant:

$$\theta_{\mathcal{M}^{(r)}}^{(next)} = \frac{1}{K} \sum_{k=0}^{K-1} \theta_{target}^{(r-k)}$$

Exponential Moving Average (EMA) Variant:

$$\theta_{\mathcal{M}^{(r)}}^{(next)} = \rho \theta_{current}^{(r)} + (1 - \rho) \theta_{target}^{(r)}$$

Pure Exponential Moving Average (Pure EMA) Variant: Utilizing a constant expansion coefficient α_{ema} independent of epoch counting:

$$\theta_{target}^{(r)} = \alpha_{ema} Var_{subspace}(\Delta^{(e^*)})$$

$$\theta_{\mathcal{M}^{(r)}}^{(next)} = \rho \theta_{current}^{(r)} + (1 - \rho) \theta_{target}^{(r)}$$

Server Optimizer Application: Following the threshold update, the active parameters are updated utilizing the designated server optimizer based on the aggregated pseudo gradient.

Case 1: Adam Server Optimizer

$$\text{Server_Optimizer(Adam)} = (W_{act}^{(r)}, \nabla_{pseudo}^{(r)}, \eta_s, \beta_1, \beta_2, \epsilon)$$

Expansion:

$$\begin{aligned}
m^{(r)} &= \beta_1 m^{(r-1)} + (1-\beta_1)\nabla_{pseudo}^{(r)} \\
v^{(r)} &= \beta_2 v^{(r-1)} + (1-\beta_2)(\nabla_{pseudo}^{(r)})^2 \\
\hat{m}^{(r)} &= \frac{m^{(r)}}{1-\beta_1^r} \\
\hat{v}^{(r)} &= \frac{v^{(r)}}{1-\beta_2^r} \\
W_{act}^{(r+1)} &= W_{act}^{(r)} - \eta_s \frac{\hat{m}^{(r)}}{\sqrt{\hat{v}^{(r)}} + \epsilon}
\end{aligned}$$

Case 2: AdamW Server Optimizer

$$\text{Server_Optimizer(AdamW)} = (W_{act}^{(r)}, \nabla_{pseudo}^{(r)}, \eta_s, \beta_1, \beta_2, \epsilon, \lambda)$$

Expansion:

$$\begin{aligned}
m^{(r)} &= \beta_1 m^{(r-1)} + (1-\beta_1)\nabla_{pseudo}^{(r)} \\
v^{(r)} &= \beta_2 v^{(r-1)} + (1-\beta_2)(\nabla_{pseudo}^{(r)})^2 \\
\hat{m}^{(r)} &= \frac{m^{(r)}}{1-\beta_1^r} \\
\hat{v}^{(r)} &= \frac{v^{(r)}}{1-\beta_2^r} \\
W_{act}^{(r+1)} &= W_{act}^{(r)} - \eta_s \left(\frac{\hat{m}^{(r)}}{\sqrt{\hat{v}^{(r)}} + \epsilon} + \lambda W_{act}^{(r)} \right)
\end{aligned}$$

Threshold Update Strategies: SMA and EMA Numerical Example

To stabilize the high frequency variance oscillations across global communication rounds, temporal smoothing techniques such as Simple Moving Average (SMA) and Exponential Moving Average (EMA) are applied to the dynamically calculated threshold. Let us define a numerical example to illustrate their mechanics.

Notation and Setup

- Assume we are evaluating the threshold for active matrix $\mathcal{M} = A$.
- Historical proposed targets for the last 3 active rounds: $\theta_{target}^{(r-2)} = 0.8$, $\theta_{target}^{(r-1)} = 1.0$, $\theta_{target}^{(r)} = 1.2$.
- Current applied threshold: $\theta_{current}^{(r)} = 0.95$.
- SMA Window Size: $K = 3$.
- EMA Smoothing Factor: $\rho = 0.8$.

1) Simple Moving Average (SMA) Logic: SMA applies a uniform weight to a fixed window of historical target proposals. It acts as a strict low pass filter over the last K steps.

$$\theta_A^{(next)} = \frac{0.8 + 1.0 + 1.2}{3} = \frac{3.0}{3} = 1.0$$

2) Exponential Moving Average (EMA) Logic: EMA applies exponentially decreasing weights to historical targets, parameterized by ρ. It favors the historical momentum state while gently folding in the new proposal.

$$\begin{aligned}\theta_A^{(next)} &= \rho\theta_{current}^{(r)} + (1-\rho)\theta_{target}^{(r)} \\ &= 0.8(0.95) + (1-0.8)(1.2) \\ &= 0.76 + 0.2(1.2) \\ &= 0.76 + 0.24 = 1.0\end{aligned}$$

Theoretical Expansion of the SGD Step

In the algorithms defined above, the local parameter update for a client i at step t during communication round r operates dynamically on the active parameter tensor W. The generic SGD rule is given by:

$$W_{i,t}^{(r)} = W_{i,t-1}^{(r)} - \eta_c \nabla \mathcal{L}_i \left(W_{i,t-1}^{(r)} \right)$$

Notation and Symbols

- $W_{i,t-1} \in \mathbb{R}^{d_{out} \times d_{in}}$: The active parameter matrix for client i at step $t-1$.
- $x \in \mathbb{R}^{d_{in}}$: The input data vector from the client's local mini batch.
- $y^* \in \mathbb{R}^{d_{out}}$: The ground truth target vector.
- $\hat{y} \in \mathbb{R}^{d_{out}}$: The model's predicted output vector.
- η_c: The local client learning rate.

Step 1: The Forward Pass The client constructs the prediction $\hat{y}$ using the active local weight matrix:

$$\hat{y} = W_{i,t-1} x$$

Step 2: Loss Computation The local empirical risk $\mathcal{L}_i$ is computed by comparing the prediction to the ground truth. Utilizing a standard Mean Squared Error (MSE) loss:

$$\mathcal{L}_i = \frac{1}{2} \|\hat{y} - y^*\|_2^2$$

Step 3: The Backward Pass To update $W_{i,t-1}$, the client calculates the gradient of the loss with respect to W:

$$\frac{\partial \mathcal{L}_i}{\partial \hat{y}} = \hat{y} - y^*$$

$$\nabla_W \mathcal{L}_i = \left(\frac{\partial \mathcal{L}_i}{\partial \hat{y}} \right) x^T = (\hat{y} - y^*) x^T$$

Step 4: The Parameter Update The client applies the computed gradient to update the active matrix, completing one SGD step:

$$W_{i,t} = W_{i,t-1} - \eta_c \nabla_W \mathcal{L}_i$$

Numerical Toy Example

Let us define a simple 2D linear mapping.

- Active Matrix $W_{t-1} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
- Input $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, Ground Truth $y^* = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$
- Learning Rate $\eta_c = 0.1$

1) Forward Pass:

$$\hat{y} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

2) Loss Computation:

$$\mathcal{L}_i = \frac{1}{2} \left\| \begin{bmatrix} 3 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\|_2^2 = \frac{1}{2} \left((1)^2 + (0)^2 \right) = 0.5$$

3) Backward Pass (Gradient Computation):

$$\frac{\partial \mathcal{L}_i}{\partial \hat{y}} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\nabla_W \mathcal{L}_i = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

4) Parameter Update:

$$W_t = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - 0.1 \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.9 & 0.8 \\ 1.0 & 1.0 \end{bmatrix}$$

This newly updated matrix W_t serves as W_{t-1} for the subsequent local gradient iteration.

Numerical Toy Example: Server Optimizers

Let us define a simple 2D linear mapping for the global aggregation block, operating at communication round $r = 1$ with zero-initialized moment buffers ($m^{(0)} = 0, v^{(0)} = 0$).

Case 1: Adam Server Optimizer

- Active Global Matrix $W_{act}^{(1)} = \begin{bmatrix} 1.0 & 1.0 \\ 1.0 & 1.0 \end{bmatrix}$
- Aggregated Pseudo Gradient $\nabla_{pseudo}^{(1)} = \begin{bmatrix} -0.2 & 0.1 \\ 0.0 & -0.1 \end{bmatrix}$
- Parameters: $\eta_s = 1.0, \beta_1 = 0.9, \beta_2 = 0.99, \epsilon = 10^{-8}$

1) Adam Format:

$$\text{Server_Optimizer(Adam)} = \left(W_{act}^{(1)}, \nabla_{pseudo}^{(1)}, 1.0, 0.9, 0.99, 10^{-8}\right)$$

2) Moment Updates and Bias Correction ($r = 1$):

$$m^{(1)} = 0.9(0) + 0.1 \begin{bmatrix} -0.2 & 0.1 \\ 0.0 & -0.1 \end{bmatrix} = \begin{bmatrix} -0.02 & 0.01 \\ 0.0 & -0.01 \end{bmatrix} \implies \hat{m}^{(1)} = \frac{m^{(1)}}{1 - 0.9^1} = \begin{bmatrix} -0.2 & 0.1 \\ 0.0 & -0.1 \end{bmatrix}$$

$$v^{(1)} = 0.99(0) + 0.01 \begin{bmatrix} 0.04 & 0.01 \\ 0.0 & 0.01 \end{bmatrix} = \begin{bmatrix} 0.0004 & 0.0001 \\ 0.0 & 0.0001 \end{bmatrix} \implies \hat{v}^{(1)} = \frac{v^{(1)}}{1 - 0.99^1} = \begin{bmatrix} 0.04 & 0.01 \\ 0.0 & 0.01 \end{bmatrix}$$

3) Global Parameter Update:

$$\begin{aligned} W_{act}^{(2)} &= \begin{bmatrix} 1.0 & 1.0 \\ 1.0 & 1.0 \end{bmatrix} - 1.0 \left(\frac{\begin{bmatrix} -0.2 & 0.1 \\ 0.0 & -0.1 \end{bmatrix}}{\sqrt{\begin{bmatrix} 0.04 & 0.01 \\ 0.0 & 0.01 \end{bmatrix}} + 10^{-8}} \right) \\ &= \begin{bmatrix} 1.0 & 1.0 \\ 1.0 & 1.0 \end{bmatrix} - 1.0 \begin{bmatrix} -1.0 & 1.0 \\ 0.0 & -1.0 \end{bmatrix} = \begin{bmatrix} 2.0 & 0.0 \\ 1.0 & 2.0 \end{bmatrix} \end{aligned}$$

Case 2: AdamW Server Optimizer

- Active Global Matrix $W_{act}^{(1)} = \begin{bmatrix} 1.0 & 1.0 \\ 1.0 & 1.0 \end{bmatrix}$
- Aggregated Pseudo Gradient $\nabla_{pseudo}^{(1)} = \begin{bmatrix} -0.2 & 0.1 \\ 0.0 & -0.1 \end{bmatrix}$
- Parameters: $\eta_s = 1.0, \beta_1 = 0.9, \beta_2 = 0.99, \epsilon = 10^{-8}, \lambda = 0.1$

1) AdamW Format:

$$\text{Server_Optimizer(AdamW)} = \left(W_{act}^{(1)}, \nabla_{pseudo}^{(1)}, 1.0, 0.9, 0.99, 10^{-8}, 0.1\right)$$

2) Global Parameter Update (utilizing identical moments $\hat{m}^{(1)}, \hat{v}^{(1)}$ from Case 1):

$$\begin{aligned} W_{act}^{(2)} &= \begin{bmatrix} 1.0 & 1.0 \\ 1.0 & 1.0 \end{bmatrix} - 1.0 \left(\begin{bmatrix} -1.0 & 1.0 \\ 0.0 & -1.0 \end{bmatrix} + 0.1 \begin{bmatrix} 1.0 & 1.0 \\ 1.0 & 1.0 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1.0 & 1.0 \\ 1.0 & 1.0 \end{bmatrix} - \begin{bmatrix} -0.9 & 1.1 \\ 0.1 & -0.9 \end{bmatrix} = \begin{bmatrix} 1.9 & -0.1 \\ 0.9 & 1.9 \end{bmatrix} \end{aligned}$$